VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Machine Learning (23CS6PCMAL)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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(Affiliated To Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Machine Learning (23CS6PCMAL)" carried out by **Eashan Jain V (1BM23CS098),** who is a bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of a Machine Learning (23CS6PCMAL) work prescribed for the said degree.

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Github Link: https://github.com/eashanjn715/ML/tree/main/ML_Lab-main
[github_link](https://github.com/eashanjn715/ML/tree/main/ML_Lab-main)

Program 1

Write a python program to import and export data using Pandas library functions

Screenshot

LAB-1: Iris Dataset:

Task-1: Load Dataset

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target
df.head()
```

~~sepal length (cm) = sepal~~

	sepal (cm) length	sepal (cm) width	petal (cm) length	petal (cm) width	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
⋮					
5	5.0	3.6	1.4	0.2	0

Columns : 5

T2] Understand the shape

```
df.shape
```

(150, 5) → (rows, columns)

Code:

```
import pandas as pd
import numpy as np
def process_storm_data():
try:
df = pd.read_csv('cyclone_mocha_data.csv', na_values=[' ', '-', 'N/A'])
#df = pd.read_excel('coastal_erosion_stats.xlsx', sheet_name='Odisha_2024')
print("--- Data Successfully Imported ---")
print(df.head())
except FileNotFoundError:
data = {
'Date': ['12.05.23', '12.05.23', '13.05.23'],
'Time_UTC': ['1800', '2100', '0000'],
'Lat': [15.1, 15.2, 15.4],
'Long': [88.8, 88.9, 89.1],
'Central_Pressure_hPa': [954, 954, 954],
'Wind_Speed_kt': [100, 100, 100],
'Grade': ['ESCS', 'ESCS', 'ESCS']
}
df = pd.DataFrame(data)
print("--- Created Sample DataFrame from Scratch ---")
df['Wind_Speed_kmh'] = df['Wind_Speed_kt'] * 1.852
9
df.to_csv('processed_storm_data.csv', index=False)
df.to_json('storm_api_output.json', orient='records', indent=4)
# Export to Excel for stakeholder reports (e.g., NDRF or IMD)
# Requires 'openpyxl' library: pip install openpyxl
df.to_excel('storm_summary_report.xlsx', index=False, sheet_name='Summary')
print("\n--- Data Successfully Exported to CSV, JSON, and Excel ---")
if __name__ == "__main__":
process_storm_data()
```

Program 2

Demonstrate various data pre-processing techniques for a given dataset

Screenshot

T3) Explore columns

```
df.columns
df.info()
```

#	column	Non-Null Count		Dtype
0	sepal length	150	non-null	float64
1	sepal width	150	non-null	"
2	petal length	150	non-null	"
3	petal width	150	non-null	"
4	species	150	non-null	int64

T4) Statistical Summary

```
df.describe()
```

	sepal length	sepal width	petal length	Petal width	species
max	7.9	4.4	6.9	2.5	2.0
min	4.3	2.0	1.0	0.1	0.0

T5) Count Each Category

```
df['species'].value_counts()
```

species	
0	50
1	50
2	50

T7) Compare species:

```
df[df['species'] == 0].head()
df[df['species'] == 1].head()
```

Do feature values differ? Yes

T8) Correlation

```
df.corr()
```

Which features are related?

petal width - species

petal length - sepal length

T9) Sort data?

```
df.sort_values(by='petal length (cm)').head()
```

What happens to species label?

Separates the species label based on their values.

11/2/26

Code:

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler, MinMaxScaler, LabelEncoder
data = {
'Age': [25, 30, 35, np.nan, 200, 22, 38, 45, 12, np.nan], # Contains NaN and Outlier (200)
'Salary': [50000, 54000, np.nan, 62000, 58000, 52000, 120000, 60000, 48000, 55000],
'Department': ['IT', 'HR', 'IT', 'Sales', 'HR', 'IT', np.nan, 'Sales', 'IT', 'HR'],
'Purchased': ['No', 'Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'No', 'No', 'Yes']
}
df = pd.DataFrame(data)
print("Original Dataset:\n", df)
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
df['Department'] = df['Department'].fillna(df['Department'].mode()[0])
le = LabelEncoder()
df['Purchased'] = le.fit_transform(df['Purchased'])
df = pd.get_dummies(df, columns=['Department'], drop_first=True)
1
5
Q1 = df['Age'].quantile(0.25)
Q3 = df['Age'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df['Age'] = np.where(df['Age'] > upper_bound, upper_bound,
np.where(df['Age'] < lower_bound, lower_bound, df['Age']))
```

Program 3

Implement Linear and Multi-Linear Regression algorithm using appropriate dataset

Screenshot

House Price Prediction

End to End Machine Learning steps:

i) Working with real data
ii) Look at the big picture
iii) Get the data, Discover
iv) and visualize the data
v) Prepare the data, select
vi) and train the model
vii) Fine tune the model

Steps involved:

Step-0) Import required libraries

Step-i) Look at the big picture

Step-ii) Get the data

Step-iii) Discover the Data (EDA)

Step-iv) Visualize the data
4.1) Distribution of House prices
4.2) Income vs House price
4.3) Correlation Heatmap
4.4) Location-based visualization

Step-v) Prepare the data
5.1) Feature Engineering
5.2) Test Train split
5.3) Feature scaling

Step-vi) Select and train a baseline model
6.1) Linear regression

Step-vii) Train Advanced Models
7.1 Decision Tree Regressor
7.2 Random Forest Regressor

Step-viii) Fine-tune the model

Step-ix) Model comparison

Analysis:

- The tuned Random Forest performed best because it achieved the lowest RMSE

Q2) Why did Random Forest outperform Linear Regression?

- Captures non-linear relationships b/w house prices & other features
- Handles feature interactions

Q3) How did tuning improve performance?

- Limiting the tree depth (max_depth)
- Adjusting how splits happen (training vs testing)
- Optimizing the iterations (n_estimators)

Multi-Linear Regression:

California Housing Dataset

Test_size : 0.3
Train_size : 0.7
Model Trained
Enter:
MedInc : 5
HouseAge : 20
Avg rooms : 6
Avg Bedrooms : 1000
Avg Occupan : 3
Latitude : 34
~~Predicted~~ Longitude : -118

Predicted value : $ 2,3 23, 465.

25/2/26

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
df_canada = pd.read_csv('canada_per_capita_income.csv')
X = df_canada[['year']]
y = df_canada['per capita income (US$)']
model = LinearRegression()
model.fit(X, y)
prediction_2020 = model.predict([[2020]])
print(f"Predicted Per Capita Income for 2020: ${prediction_2020[0]:.2f}")
import pandas as pd
from sklearn.linear_model import LinearRegression
df_salary = pd.read_csv('salary.csv')
df_salary['YearsExperience'] =
df_salary['YearsExperience'].fillna(df_salary['YearsExperience'].median())
X_salary = df_salary[['YearsExperience']]
y_salary = df_salary['Salary']
model_salary = LinearRegression()
model_salary.fit(X_salary, y_salary)
predicted_salary = model_salary.predict([[12]])
print(f"Predicted Salary for 12 years of experience: ${predicted_salary[0]:.2f}")
```

Program 4

Build Logistic Regression Model for a given dataset

Screenshot

Logistic Regression

HR_comma_sep.csv

Model Accuracy: 0.766

Confusion matrix:

$$\begin{bmatrix} 2118 & 176 \\ 533 & 173 \end{bmatrix}$$

Classification Report:

	Precision	Recall	f1-score	support
0	0.8	0.92	0.86	2294
1	0.5	0.25	0.33	706

Q1) Which variables did matter most?

- Satisfaction_level
  - $\propto \frac{1}{\text{retention}}$
- Average_monthly_hours
- work_accident
- Salary

Q2) What was the accuracy?

Accuracy = 76.6%

It is good because data is real world and not ideal.

ZOO - Dataset.

Accuracy: 0.9523

confusion matrix:

$$\begin{pmatrix} 12 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Report

	Precision	Recall	F1	Support
1	1.00	1	1.00	12
2	1.00	1	1.00	2
3	0.80	0	0.00	1
4	0.67	1	0.80	2

Q1] Did you perform any pre-processing? why?

- Encoding variables
- Feature selection
- Scaling

Q2] Data had no missing values & was clean & structured

- Confusion matrix shows
  - Correct predictions → Diagonal
  - Misclassifications → Non-Diagonal

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
# 1. Load the dataset and Preprocessing
# Note: Ensure the csv files are in your working directory
try:
zoo_df = pd.read_csv('zoo-data.csv')
# Preprocessing:
# Usually, 'animal_name' is a unique identifier and should be dropped
# as it provides no predictive value.
if 'animal_name' in zoo_df.columns:
zoo_df = zoo_df.drop('animal_name', axis=1)
# Separate features (X) and target (y)
X = zoo_df.drop('class_type', axis=1)
y = zoo_df['class_type']
# Split into Training (80%) and Testing (20%) sets
2
9
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# 2. Build Logistic Regression Model
# We use multi_class='multinomial' for targets with more than two categories
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
# 3. Measure the accuracy
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
# 4. Plot the Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(10, 7))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted Class')
plt.ylabel('Actual Class')
plt.title('Confusion Matrix - Zoo Animal Classification')
plt.show()
3
0
except FileNotFoundError:
print("Error: Please ensure 'zoo-data.csv' is in the same folder as this script.")
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
# 1. Load the dataset and Preprocessing
# Note: Ensure the csv files are in your working directory
try:
zoo_df = pd.read_csv('zoo-data.csv')
# Preprocessing:
# Usually, 'animal_name' is a unique identifier and should be dropped
# as it provides no predictive value.
if 'animal_name' in zoo_df.columns:
zoo_df = zoo_df.drop('animal_name', axis=1)
# Separate features (X) and target (y)
3
1
X = zoo_df.drop('class_type', axis=1)
y = zoo_df['class_type']
# Split into Training (80%) and Testing (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# 2. Build Logistic Regression Model
# We use multi_class='multinomial' for targets with more than two categories
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
# 3. Measure the accuracy
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
# 4. Plot the Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(10, 7))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted Class')
3
2
plt.ylabel('Actual Class')
plt.title('Confusion Matrix - Zoo Animal Classification')
plt.show()
except FileNotFoundError:
print("Error: Please ensure 'zoo-data.csv' is in the same folder as this script.")
```

Program 5

Use an appropriate data set for building the decision tree (ID3) and apply this knowledge to classify a new sample.

Screenshot

LAB 4:

ID3

Decision Tree: (Iris Dataset)

Petal length ≤ 2.45

class: 0

Petal length > 2.45

petal length ≤ 4.75

petal length > 4.75

petal width ≤ 1.60

petal width > 1.60

petal length ≤ 1.5

petal ≥ 5.15

class: 1

class: 2

class: 2

class: 2

Output:

Classification Accuracy: 0.977

class	Precision	Recall	F1	Support
0	1.00	1.00	1.00	19
1	1.00	0.92	0.96	13
2	0.93	1.00	0.96	13

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.preprocessing import LabelEncoder
def build_classification_model(file_path, target_column):
df = pd.read_csv(file_path)
le = LabelEncoder()
for col in df.columns:
X = df.drop(target_column, axis=1)
y = df[target_column]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
classifier = DecisionTreeClassifier(criterion='entropy')
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
print(f"--- Results for {file_path} ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred):.4f}")
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
try:
build_classification_model('iris.csv', 'species')
build_classification_model('drug.csv', 'Drug')
except FileNotFoundError as e:
print(f"Error: {e}")
import pandas as pd
import numpy as np
3
8
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
try:
petrol_df = pd.read_csv('petrol_consumption.csv')
X = petrol_df.drop('Petrol_Consumption', axis=1)
y = petrol_df['Petrol_Consumption']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
regressor = DecisionTreeRegressor()
regressor.fit(X_train, y_train)
y_pred = regressor.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
print("--- Results for Petrol Consumption Regression ---")
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")
except FileNotFoundError:
print("Error: petrol_consumption.csv not found.")
```

Program 6

Build KNN Classification model for a given dataset.

Screenshot

Implementation of KNN:

Algorithm:

1. Assign a value for K, usually ranges from (3-5)
2. Consider the values in the given dataset (given in key-value pairs)
3. Compute Euclidean distance of each point with the point considered, using:

$$d = \sqrt{(x-x_i)^2 + (y-y_i)^2 + \ldots}$$

4. Sort the distance in ascending order
5. Choose K-nearest neighbours from first.
6. Classify the data point based on majority of K points considered.

Example: Iris dataset

Output:

Actual:	Predicted
versicolor	versicolor
setosa	setosa
versicolor	versicolor
virginia	~~versicolor~~ virginia

Accuracy: 1.0.

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, classification_report, accuracy_score
iris_df = pd.read_csv('iris.csv')
X = iris_df.drop('species', axis=1)
y = iris_df['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
print("--- IRIS Dataset Results ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred)}")
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
diabetes_df = pd.read_csv('diabetes.csv')
cols_fix = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
for col in cols_fix:
diabetes_df[col] = diabetes_df[col].replace(0, diabetes_df[col].median())
X_d = diabetes_df.drop('Outcome', axis=1)
4
2
y_d = diabetes_df['Outcome']
X_train_d, X_test_d, y_train_d, y_test_d = train_test_split(X_d, y_d, test_size=0.2,
random_state=42)
scaler_d = StandardScaler()
X_train_d = scaler_d.fit_transform(X_train_d)
X_test_d = scaler_d.transform(X_test_d)
knn_d = KNeighborsClassifier(n_neighbors=11) #
knn_d.fit(X_train_d, y_train_d)
y_pred_d = knn_d.predict(X_test_d)
print("\n--- Diabetes Dataset Results ---")
print(f"Accuracy Score: {accuracy_score(y_test_d, y_pred_d)}")
print("Confusion Matrix:\n", confusion_matrix(y_test_d, y_pred_d))
import matplotlib.pyplot as plt
import seaborn as sns
heart_df = pd.read_csv('heart.csv')
X_h = heart_df.drop('target', axis=1)
y_h = heart_df['target']
X_train_h, X_test_h, y_train_h, y_test_h = train_test_split(X_h, y_h, test_size=0.2,
random_state=42)
scaler_h = StandardScaler()
X_train_h = scaler_h.fit_transform(X_train_h)
```

```
X_test_h = scaler_h.transform(X_test_h)
scores = []
for k in range(1, 21):
knn_h = KNeighborsClassifier(n_neighbors=k)
knn_h.fit(X_train_h, y_train_h)
scores.append(knn_h.score(X_test_h, y_test_h))
```
4
3
```
best_k = scores.index(max(scores)) + 1
print(f"\nBest K value found: {best_k} with score: {max(scores)}")
final_knn = KNeighborsClassifier(n_neighbors=best_k)
final_knn.fit(X_train_h, y_train_h)
y_pred_h = final_knn.predict(X_test_h)
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test_h, y_pred_h), annot=True, fmt='d', cmap='Blues')
plt.title(f'Confusion Matrix (K={best_k})')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
report = classification_report(y_test_h, y_pred_h, output_dict=True)
sns.heatmap(pd.DataFrame(report).iloc[:-1, :].T, annot=True, cmap='YlGnBu')
plt.title('Classification Report')
plt.show()
```

Program 7

Build Support vector machine model for a given dataset.

Screenshot

SVM:

Accuracy: 0.799

Confusion matrix:

$$\begin{bmatrix} 106 & 7 \\ 35 & 61 \end{bmatrix}$$

Classification report:

	Precision	Recall	F1-score	support
0	0.75	0.94	0.83	113
1	0.90	0.64	0.74	96
Accuracy	0.82	0.79	0.79	209

Boosting

Accuracy: 0.933

Classification:

	Precision	Recall	F1-score	support
0	1.00	1.00	1.00	10
1	0.89	0.89	0.89	9
2	0.91	0.91	0.91	11
Accuracy	0.93	0.93	0.93	30

0/4/26

confusion matrix:

$$\begin{bmatrix} 10 & 0 & 0 \\ 0 & 8 & 1 \\ 0 & 1 & 10 \end{bmatrix}$$

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix
df = pd.read_csv('iris.csv')
X = df.drop('species', axis=1)
y = df['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
svm_linear = SVC(kernel='linear')
svm_linear.fit(X_train, y_train)
y_pred_linear = svm_linear.predict(X_test)
print("--- Linear Kernel Results ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred_linear)}")
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_linear))
svm_rbf = SVC(kernel='rbf')
svm_rbf.fit(X_train, y_train)
y_pred_rbf = svm_rbf.predict(X_test)
4
9
print("\n--- RBF Kernel Results ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred_rbf)}")
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rbf))
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, roc_curve, auc
from sklearn.preprocessing import label_binarize, LabelEncoder
df_letter = pd.read_csv('letter-recognition.csv')
X_l = df_letter.drop('letter', axis=1)
y_l = df_letter['letter']
le = LabelEncoder()
y_enc = le.fit_transform(y_l)
n_classes = len(le.classes_)
X_train_l, X_test_l, y_train_l, y_test_l = train_test_split(X_l, y_enc, test_size=0.2,
random_state=42)
5
0
# 3. Build SVM Classifier (using probability=True for ROC/AUC)
svm_model = SVC(probability=True, random_state=42)
svm_model.fit(X_train_l, y_train_l)
# 4. Predictions and Scores
y_pred_l = svm_model.predict(X_test_l)
y_score_l = svm_model.predict_proba(X_test_l)
```

```
print(f"Accuracy Score: {accuracy_score(y_test_l, y_pred_l)}")
print("Confusion Matrix (Partial view):")
print(confusion_matrix(y_test_l, y_pred_l))
# 5. ROC Curve and AUC Score (Micro-average for Multiclass)
y_test_bin = label_binarize(y_test_l, classes=np.arange(n_classes))
fpr, tpr, _ = roc_curve(y_test_bin.ravel(), y_score_l.ravel())
roc_auc = auc(fpr, tpr)
print(f"Micro-average AUC Score: {roc_auc}")
# 6. Plotting
plt.figure()
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area = {roc_auc:.4f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
```



```
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve: Letter Recognition')
plt.legend(loc="lower right")
plt.show()
```

Program 8

Implement Random forest ensemble method on a given dataset.

Screenshot

LAB-6:

Random Forest:

- Titanic dataset

	pclass	sex	age	fare	survived
0	1	1	29.00	211.33	1
1	1	0	0.916	151.55	1
2	1	1	2.00	151.55	0
3	1	0	30.00	151.55	0
4	1	1	25.00	151.55	0

Accuracy: 0.81297

Classification report:

	Precision	Recall	F1 score	Support
0	0.79	0.9	0.84	144
1	0.85	0.71	0.77	118
Accuracy			0.81	262
macro avg	0.82	0.80	0.81	262
weighted avg	0.82	0.81	0.81	262

Confusion Matrix:

$$\begin{bmatrix} 129 & 15 \\ 34 & 84 \end{bmatrix}$$

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
df = pd.read_csv('iris.csv')
X = df.drop('species', axis=1)
y = df['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
rf_default = RandomForestClassifier(n_estimators=10, random_state=42)
rf_default.fit(X_train, y_train)
y_pred_default = rf_default.predict(X_test)
score_10 = accuracy_score(y_test, y_pred_default)
print(f"Accuracy with n_estimators=10: {score_10 * 100:.2f}%")
best_score = 0
best_n = 0
for n in range(1, 101):
5
6
rf = RandomForestClassifier(n_estimators=n, random_state=42)
rf.fit(X_train, y_train)
score = rf.score(X_test, y_test)
if score > best_score:
best_score = score
best_n = n
print(f"Best Accuracy Score: {best_score * 100:.2f}%")
print(f"Number of trees for best score: {best_n}")
```

Program 9

Implement Boosting ensemble method on a given dataset.

Screenshot

SVM:

Accuracy: 0.799

Confusion matrix:

$$\begin{bmatrix} 106 & 7 \\ 35 & 61 \end{bmatrix}$$

Classification report:

	Precision	Recall	F1-score	support
0	0.75	0.94	0.83	113
1	0.90	0.64	0.74	96
Accuracy	0.82	0.79	0.79	209

Boosting

Accuracy: 0.933

Classification:

	Precision	Recall	F1-score	support
0	1.00	1.00	1.00	10
1	0.89	0.89	0.89	9
2	0.91	0.91	0.91	11
Accuracy	0.93	0.93	0.93	30

8

0/4/26

confusion matrix:

$$\begin{bmatrix} 10 & 0 & 0 \\ 0 & 8 & 1 \\ 0 & 1 & 10 \end{bmatrix}$$

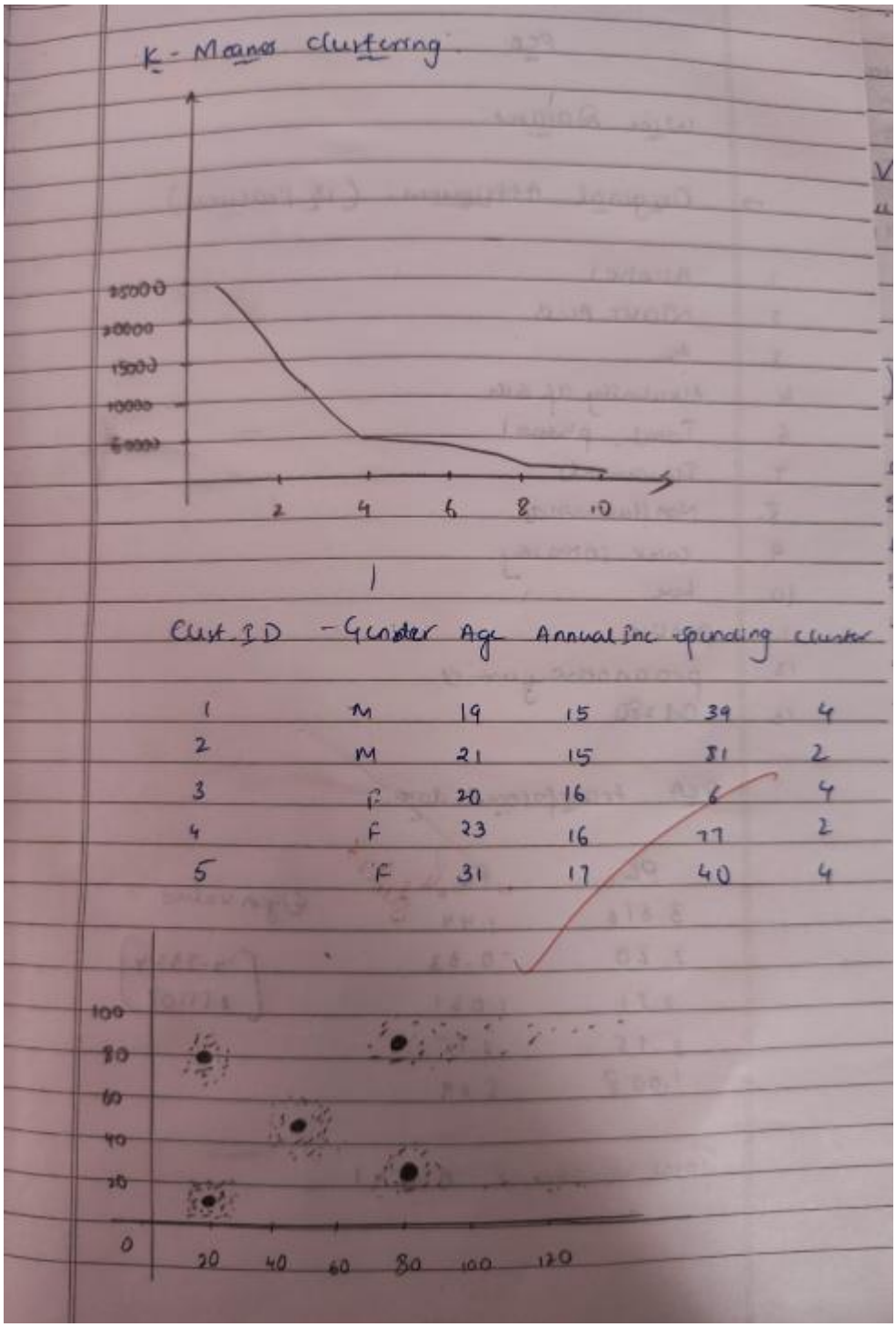
Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score
df = pd.read_csv('income.csv')
# Separate features (X) and target (y)
X = df.drop('income_level', axis=1)
y = df['income_level']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
ada_10 = AdaBoostClassifier(n_estimators=10, random_state=42)
ada_10.fit(X_train, y_train)
y_pred_10 = ada_10.predict(X_test)
score_10 = accuracy_score(y_test, y_pred_10)
print(f"Prediction score with 10 trees: {score_10:.4f}")
# Testing various numbers of trees to identify the best performance
tree_counts = [5, 10, 25, 50, 100, 150, 200, 300, 400, 500]
scores = []
print("\nFine-tuning results:")
for n in tree_counts:
model = AdaBoostClassifier(n_estimators=n, random_state=42)
model.fit(X_train, y_train)
acc = model.score(X_test, y_test)
scores.append(acc)
print(f"Trees: {n:3} | Accuracy: {acc:.4f}")
best_acc = max(scores)
best_n = tree_counts[scores.index(best_acc)]
6
1
print(f"\nOptimal performance: {best_acc:.4f} accuracy achieved with {best_n} trees.")
# 5. Visualization (Optional but helpful)
plt.figure(figsize=(8, 5))
plt.plot(tree_counts, scores, marker='o', linestyle='-', color='darkgreen')
plt.title('AdaBoost Accuracy vs. Number of Trees')
plt.xlabel('n_estimators')
plt.ylabel('Test Accuracy')
plt.grid(True)
plt.show()
```

Program 10

Build k-Means algorithm to cluster a set of data stored in a .CSV file.

Screenshot



K - Means clustering

Cust. ID	Gender	Age	Annual Inc	Spending	cluster
1	M	19	15	39	4
2	M	21	15	81	2
3	F	20	16	6	4
4	F	23	16	77	2
5	F	31	17	40	4

Code:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA
df = pd.read_csv('heart.csv')
categorical_features = ['cp', 'restecg', 'slope', 'ca', 'thal']
df_encoded = pd.get_dummies(df, columns=categorical_features, drop_first=True)
X = df_encoded.drop('target', axis=1)
y = df_encoded['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
models = {
"Logistic Regression": LogisticRegression(random_state=42),
"SVM": SVC(random_state=42),
"Random Forest": RandomForestClassifier(random_state=42)
}
print("--- Accuracies Before PCA ---")
results_before = {}
for name, model in models.items():
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
acc = accuracy_score(y_test, y_pred)
results_before[name] = acc
print(f"{name}: {acc:.4f}")
pca = PCA(n_components=0.95)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
print(f"\nOriginal dimensions: {X_train_scaled.shape[1]}")
print(f"Dimensions after PCA: {X_train_pca.shape[1]}")
print("\n--- Accuracies After PCA ---")
results_after = {}
for name, model in models.items():
model.fit(X_train_pca, y_train)
y_pred = model.predict(X_test_pca)
acc = accuracy_score(y_test, y_pred)
results_after[name] = acc
print(f"{name}: {acc:.4f}")
```

Program 10

Implement Dimensionality reduction using Principal Component Analysis (PCA) method.

Screenshot

PCA:

Wine Dataset:

→ Original Attributes: (13 features)

1. Alcohol
2. Malic Acid
3. Ash
4. Alkalinity of ash
6. Total phenol
7. Flavanoids
8. Nonflavanoids
9. color intensity
10. hue
11. proline
12. proanthocyanins
13. Od 280

PCA transformed data

PC_1	PC_2
3.316	1.44
2.20	-0.33
2.51	1.031
3.75	2.75
1.008	8.69

Eigen values

$$\begin{bmatrix} 4.7324 \\ 2.51108 \end{bmatrix}$$

Total variance = 0.5541

Code:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA
df = pd.read_csv('heart.csv')
categorical_cols = ['cp', 'restecg', 'slope', 'ca', 'thal']
df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
X = df_encoded.drop('target', axis=1)
y = df_encoded['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
7
4
def evaluate_models(X_tr, X_te, y_tr, y_te):
models = {
"SVM": SVC(),
"Logistic Regression": LogisticRegression(),
"Random Forest": RandomForestClassifier(random_state=42)
}
accuracies = {}
for name, model in models.items():
model.fit(X_tr, y_tr)
y_pred = model.predict(X_te)
accuracies[name] = accuracy_score(y_te, y_pred)
return accuracies
results_initial = evaluate_models(X_train_scaled, X_test_scaled, y_train, y_test)
print("--- Accuracies Without PCA ---")
for model, acc in results_initial.items():
print(f"{model}: {acc:.4f}")
pca = PCA(n_components=0.95)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
print(f"\nOriginal dimensions: {X_train_scaled.shape[1]}")
7
5
print(f"Dimensions after PCA: {X_train_pca.shape[1]}")
results_pca = evaluate_models(X_train_pca, X_test_pca, y_train, y_test)
print("\n--- Accuracies With PCA ---")
for model, acc in results_pca.items():
print(f"{model}: {acc:.4f}")
```